

$$h := 12 \text{ m}$$

$$l := 0.8 \cdot h = 9.6 \text{ m}$$

1-determination of the critical failure surface
by using linear interpolation

$$\omega := 39.1 \text{ deg}$$

$$k_c := 0.22$$

$$k_q := 0.62$$

$$D := 4.584 \text{ m}$$

$$\varphi := 29 \text{ deg}$$

$$\varepsilon := 90 \text{ deg} - (\omega + \varphi) = 21.9 \text{ deg}$$

to get the values for the exponential function drawing

$$r_A := 13.84 \text{ m}$$

$$r_B := 20.165 \text{ m}$$

$$\omega_1 := 5 \text{ deg}$$

$$\omega_2 := 10 \text{ deg}$$

$$\omega_3 := 15 \text{ deg}$$

$$\omega_4 := 20 \text{ deg}$$

$$\omega_5 := 25 \text{ deg}$$

$$\omega_6 := 30 \text{ deg}$$

$$\omega_7 := 35 \text{ deg}$$

$$r_i := r_A \cdot e^{\omega_i \cdot \tan(\phi)}$$

$$r := 1452.5933 \text{ cm}$$

$$r_5 := 1762.6896 \text{ cm}$$

$$r_2 := 1524.5862 \text{ cm}$$

$$r_6 := 1850.0514 \text{ cm}$$

$$r_3 := 1600.1472 \text{ cm}$$

$$r_7 := 1941.7429 \text{ cm}$$

$$r_4 := 1679.4531 \text{ cm}$$

2-calculation of earth pressure parameters

$$k_{cs} := 0.6194$$

$$\sigma_{hAf} := 36.456 \text{ kPa}$$

$$\sigma_{hBf} := 22.32 \text{ kPa}$$

$$H_0 := 4.2623 \text{ m}$$

$$\sigma_{hAH0} := 18.7542 \text{ kPa}$$

$$\sigma_{hBH0} := 22.5051 \text{ kPa}$$

calculation of horizontale earth pressure

$$\sigma_{h1} := 34.8708 \text{ kPa}$$

$$\sigma_{h5} := 22.499 \text{ kPa}$$

$$\sigma_{h9} := 22.5051 \text{ kPa}$$

$$\sigma_{h13} := 22.5051 \text{ kPa}$$

$$\sigma_{h2} := 31.7004 \text{ kPa}$$

$$\sigma_{h6} := 22.5051 \text{ kPa}$$

$$\sigma_{h10} := 22.5051 \text{ kPa}$$

$$\sigma_{h14} := 22.5051 \text{ kPa}$$

$$\sigma_{h3} := 28.53 \text{ kPa}$$

$$\sigma_{h7} := 22.5051 \text{ kPa}$$

$$\sigma_{h11} := 22.5051 \text{ kPa}$$

$$\sigma_{h15} := 22.5051 \text{ kPa}$$

$$\sigma_{h4} := 25.3596 \text{ kPa}$$

$$\sigma_{h8} := 22.5051 \text{ kPa}$$

$$\sigma_{h12} := 22.5051 \text{ kPa}$$

$$\sigma_{h16} := 22.5051 \text{ kPa}$$

calculation of vertical earth pressure

$$\sigma_{v1} := 37.5 \text{ kPa}$$

$$\sigma_{v5} := 97.5 \text{ kPa}$$

$$\sigma_{v9} := 157.5 \text{ kPa}$$

$$\sigma_{v13} := 217.5 \text{ kPa}$$

$$\sigma_{v2} := 52.5 \text{ kPa}$$

$$\sigma_{v6} := 112.5 \text{ kPa}$$

$$\sigma_{v10} := 172.5 \text{ kPa}$$

$$\sigma_{v14} := 232.5 \text{ kPa}$$

$$\sigma_{v3} := 67.5 \text{ kPa}$$

$$\sigma_{v7} := 127.5 \text{ kPa}$$

$$\sigma_{v11} := 187.5 \text{ kPa}$$

$$\sigma_{v15} := 247.5 \text{ kPa}$$

$$\sigma_{v4} := 82.5 \text{ kPa}$$

$$\sigma_{v8} := 142.5 \text{ kPa}$$

$$\sigma_{v12} := 202.5 \text{ kPa}$$

$$\sigma_{v16} := 262.5 \text{ kPa}$$

calculation of the normal forces on each strip (charctaristic values)

$N_{k1} := 19.6148 \text{ kN}$	$N_{k5} := 12.6411 \text{ kN}$	$N_{k9} := 12.6591 \text{ kN}$	$N_{k13} := 12.6591 \text{ kN}$
$N_{k2} := 17.8315 \text{ kN}$	$N_{k6} := 12.6557 \text{ kN}$	$N_{k10} := 12.6591 \text{ kN}$	$N_{k14} := 12.6591 \text{ kN}$
$N_{k3} := 16.0481 \text{ kN}$	$N_{k7} := 12.6591 \text{ kN}$	$N_{k11} := 12.6591 \text{ kN}$	$N_{k15} := 12.6591 \text{ kN}$
$N_{k4} := 14.2648 \text{ kN}$	$N_{k8} := 12.6591 \text{ kN}$	$N_{k12} := 12.6591 \text{ kN}$	$N_{k16} := 12.6591 \text{ kN}$

calculation of the normal forces on each strip (design values)

$N_{d1} := 26.48 \text{ kN}$	$N_{d5} := 17.0654 \text{ kN}$	$N_{d9} := 17.0898 \text{ kN}$	$N_{d13} := 17.0898 \text{ kN}$
$N_{d2} := 24.0725 \text{ kN}$	$N_{d6} := 17.0852 \text{ kN}$	$N_{d10} := 17.0898 \text{ kN}$	$N_{d14} := 17.0898 \text{ kN}$
$N_{d3} := 21.665 \text{ kN}$	$N_{d7} := 17.0898 \text{ kN}$	$N_{d11} := 17.0898 \text{ kN}$	$N_{d15} := 17.0898 \text{ kN}$
$N_{d4} := 19.2575 \text{ kN}$	$N_{d8} := 17.0898 \text{ kN}$	$N_{d12} := 17.0898 \text{ kN}$	$N_{d16} := 17.0898 \text{ kN}$

calculation of the width of each strip

$s_1 := 0.2 \text{ m}$	$s_5 := 0.049 \text{ m}$	$s_9 := 0.026 \text{ m}$	$s_{13} := 0.015 \text{ m}$
$s_2 := 0.136 \text{ m}$	$s_6 := 0.041 \text{ m}$	$s_{10} := 0.023 \text{ m}$	$s_{14} := 0.013 \text{ m}$
$s_3 := 0.094 \text{ m}$	$s_7 := 0.035 \text{ m}$	$s_{11} := 0.02 \text{ m}$	$s_{15} := 0.012 \text{ m}$
$s_4 := 0.067 \text{ m}$	$s_8 := 0.03 \text{ m}$	$s_{12} := 0.017 \text{ m}$	$s_{16} := 0.01 \text{ m}$

calculation of the applied width of each strip

$s_{apl.1} := 20 \text{ cm}$	$s_{apl.5} := 6 \text{ cm}$	$s_{apl.9} := 4 \text{ cm}$	$s_{apl.13} := 4 \text{ cm}$
$s_{apl.2} := 15 \text{ cm}$	$s_{apl.6} := 6 \text{ cm}$	$s_{apl.10} := 4 \text{ cm}$	$s_{apl.14} := 4 \text{ cm}$
$s_{apl.3} := 10 \text{ cm}$	$s_{apl.7} := 4 \text{ cm}$	$s_{apl.11} := 4 \text{ cm}$	$s_{apl.15} := 4 \text{ cm}$
$s_{apl.4} := 8 \text{ cm}$	$s_{apl.8} := 4 \text{ cm}$	$s_{apl.12} := 4 \text{ cm}$	$s_{apl.16} := 4 \text{ cm}$

calculation for the pull out test

$K_1 := 1.42$	$K_5 := 1.81$	$K_9 := 2.28$	$K_{13} := 3.92$
$K_2 := 1.64$	$K_6 := 2.15$	$K_{10} := 2.63$	$K_{14} := 4.45$
$K_3 := 1.57$	$K_7 := 1.69$	$K_{11} := 3.01$	$K_{15} := 5.04$
$K_4 := 1.76$	$K_8 := 1.97$	$K_{12} := 3.44$	$K_{16} := 5.7$

calculation for the lenght of the strip in the passive zone

$D_1 := 4.59 \text{ m}$	$D_2 := 4.59 \text{ m}$	$D_3 := 4.54 \text{ m}$	$D_4 := 4.45 \text{ m}$	$D_5 := 4.32 \text{ m}$	$D_6 := 4.15 \text{ m}$
$D_7 := 3.94 \text{ m}$	$D_8 := 3.695 \text{ m}$	$D_9 := 3.41 \text{ m}$	$D_{10} := 3.085 \text{ m}$	$D_{11} := 2.725 \text{ m}$	$D_{12} := 2.33 \text{ m}$
$D_{13} := 1.89 \text{ m}$	$D_{14} := 1.415 \text{ m}$	$D_{15} := 0.89 \text{ m}$	$D_{16} := 0.315 \text{ m}$		

$(l - D_1) = 5.01 \text{ m}$	$(l - D_5) = 5.28 \text{ m}$	$(l - D_9) = 6.19 \text{ m}$	$(l - D_{13}) = 7.71 \text{ m}$
$(l - D_2) = 5.01 \text{ m}$	$(l - D_6) = 5.45 \text{ m}$	$(l - D_{10}) = 6.515 \text{ m}$	$(l - D_{14}) = 8.185 \text{ m}$
$(l - D_3) = 5.06 \text{ m}$	$(l - D_7) = 5.66 \text{ m}$	$(l - D_{11}) = 6.875 \text{ m}$	$(l - D_{15}) = 8.71 \text{ m}$
$(l - D_4) = 5.15 \text{ m}$	$(l - D_8) = 5.905 \text{ m}$	$(l - D_{12}) = 7.27 \text{ m}$	$(l - D_{16}) = 9.285 \text{ m}$

$S_{apl.sum} := 1.05 \text{ m}$
$l_{sum} := 103.265 \text{ m}$
$K_n := 95.8911$

checking if $N_{id} \geq N_{pl.rd}$

$N_{pl.Rd1} := 26.48 \text{ kN}$	$N_{pl.Rd5} := 17.0654 \text{ kN}$	$N_{pl.Rd9} := 17.0898 \text{ kN}$	$N_{pl.Rd13} := 17.0898 \text{ kN}$
$N_{pl.Rd2} := 24.0725 \text{ kN}$	$N_{pl.Rd6} := 17.0852 \text{ kN}$	$N_{pl.Rd10} := 17.0898 \text{ kN}$	$N_{pl.Rd14} := 17.0898 \text{ kN}$
$N_{pl.Rd3} := 21.665 \text{ kN}$	$N_{pl.Rd7} := 17.0898 \text{ kN}$	$N_{pl.Rd11} := 17.0898 \text{ kN}$	$N_{pl.Rd15} := 17.0898 \text{ kN}$
$N_{pl.Rd4} := 19.2575 \text{ kN}$	$N_{pl.Rd8} := 17.0898 \text{ kN}$	$N_{pl.Rd12} := 17.0898 \text{ kN}$	$N_{pl.Rd16} := 17.0898 \text{ kN}$

from here we can see that they're equal in every strip

calculations for the thickness of the thickness of the strips

$v_1 := 0.48 \text{ mm}$	$v_5 := 1.03 \text{ mm}$	$v_9 := 1.55 \text{ mm}$	$v_{13} := 1.55 \text{ mm}$
$v_2 := 0.58 \text{ mm}$	$v_6 := 1.04 \text{ mm}$	$v_{10} := 1.55 \text{ mm}$	$v_{14} := 1.55 \text{ mm}$
$v_3 := 0.88 \text{ mm}$	$v_7 := 1.55 \text{ mm}$	$v_{11} := 1.55 \text{ mm}$	$v_{15} := 1.55 \text{ mm}$
$v_4 := 0.98 \text{ mm}$	$v_8 := 1.55 \text{ mm}$	$v_{12} := 1.55 \text{ mm}$	$v_{16} := 1.55 \text{ mm}$

calculations for the thickness of the thickness increasing of the strips due to corrosion

$v_k := 5.4928$

calculation of the applied thickness for each strip

$v_{apl.1} := 5.9743 \text{ mm}$	$v_{apl.5} := 6.5271 \text{ mm}$	$v_{apl.9} := 7.0464 \text{ mm}$	$v_{apl.13} := 7.0464 \text{ mm}$
$v_{apl.2} := 6.0764 \text{ mm}$	$v_{apl.6} := 6.5283 \text{ mm}$	$v_{apl.10} := 7.0464 \text{ mm}$	$v_{apl.14} := 7.0464 \text{ mm}$
$v_{apl.3} := 6.3682 \text{ mm}$	$v_{apl.7} := 7.0464 \text{ mm}$	$v_{apl.11} := 7.0464 \text{ mm}$	$v_{apl.15} := 7.0464 \text{ mm}$
$v_{apl.4} := 6.4776 \text{ mm}$	$v_{apl.8} := 7.0464 \text{ mm}$	$v_{apl.12} := 7.0464 \text{ mm}$	$v_{apl.16} := 7.0464 \text{ mm}$

3-analysis of sliding

$$\beta := 0 \quad \alpha := 0 \quad \delta := \varphi$$

$$K_a := \frac{\cos(\delta) \cdot (\cos(\varphi + \alpha))^2}{(\cos(\alpha))^2 \cdot \left(\sqrt{\cos(\alpha - \delta)} + \sqrt{\frac{\sin(\varphi + \delta) \cdot \sin(\varphi - \beta)}{\cos(\alpha + \beta)}} \right)^2} = 0.2692$$

$$P_{a.g.h.k} := 387.6854 \frac{\text{kN}}{\text{m}}$$

$$P_{a.q.h.k} := 77.5371 \frac{\text{kN}}{\text{m}}$$

$$E_d := 628.0503 \frac{\text{kN}}{\text{m}}$$

if we have to work with cohesive soil surface which is clay

$$R_d := 698.1818 \frac{\text{kN}}{\text{m}}$$

if case we have to work with granular soil surface

$$G_K := 240 \frac{\text{kN}}{\text{m}}$$

$$\tan(\delta_b) := \tan(\delta)$$

$$R_d := (G_K) \cdot \frac{\tan(\delta_b)}{1.1} = 120.9402 \frac{\text{kN}}{\text{m}}$$

$$\Lambda := \frac{E_d}{R_d} = 5.1931$$

retaining wall is not sufficiently secured against sliding.

4-analysis of bearing capacity

$$R_K := 411.3274 \text{ kPa}$$

$$R_d := \frac{R_K}{1.4} = 293.8053 \text{ kPa}$$

$$g_d := 30 \frac{\text{kN}}{\text{m}} \cdot 1.35 = 40.5 \text{ kPa}$$

$$g_d := 24 \frac{\text{kN}}{\text{m}} \cdot 1.35 = 32.4 \text{ kPa}$$

$$V_d := 396.9 \text{ kPa}$$

not fulfilled since $R_d < V_d$

$$h_{max} := 8.1817 \text{ m}$$